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Quandela Challenge
ETH Quantum Hackathon 2026

Are Quantum PINNs Actually Doing Anything?

Physics-Informed Neural Networks (PINNs) are a powerful approach for solving partial differential equations (PDEs), where instead of relying purely on data, a neural network is trained to satisfy the underlying physical laws of a system. By embedding the governing equations directly into the loss function, PINNs can learn solutions that are consistent with physics, even in settings where data is scarce or noisy.

Recently, there has been growing interest in combining PINNs with quantum models. The idea is appealing: quantum circuits might offer richer function representations, different inductive biases, or advantages in high-dimensional problems. In particular, the paper "Quantum physics informed neural networks for multi-variable PDEs" suggests that quantum-enhanced models could bring something genuinely new to the table.

However, this promise comes with important open questions. The results are difficult to reproduce, comparisons to classical methods are not always clear, and many proposed quantum models operate in regimes that can still be efficiently simulated classically.

This challenge is built around a simple but important question:

Is there actually something interesting here, or is it mostly theoretical decoration?

Rather than taking the claims at face value, you will investigate them directly – by reproducing, testing, and ultimately questioning the role of the quantum component itself.

Resources: *Challenger Folder – Challenge Resources*

Phase 1 — Reproduction

The first step is to establish a reliable baseline and verify the core claims of the paper under controlled conditions. Participants will use Merlin to explore that question from several angles, starting with a grounded attempt at reproduction. The goal is to reimplement the paper as faithfully as possible, capturing its core ideas in a controlled setting and testing how well they hold up.

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You will compare a classical PINN and a quantum-inspired version under similar parameter budgets, looking at accuracy, training behavior, and how performance scales as you increase circuit depth. If the quantum model provides an advantage, it should become visible here. If it does not, that is already a meaningful and valuable result.

From there, the focus shifts to something more subtle: whether the "quantum" model is actually quantum in any meaningful sense. Many photonic and continuous-variable circuits can be efficiently simulated classically, especially at low depth or within Gaussian regimes. Part of the challenge is therefore to understand what class your implementation falls into. If it turns out to be classically simulable, then the claimed advantage becomes much less convincing – and that is precisely the kind of insight we are looking for.

Phase 2 — Going Beyond

Once a baseline is established, you will move beyond reproduction and begin to question and redesign the model itself. Instead of reproducing a possibly arbitrary circuit, you will design something that makes physical sense within the Merlin framework. The goal is not to build the most complex architecture, but to construct a circuit that has a clear interpretation and a justified structure. This is where you can move beyond critique and propose better alternatives.

Another important direction is to question the architectural choices made in the paper. In particular, the use of an auxiliary derivative formulation – introduced to avoid computing second derivatives directly – also introduces additional complexity. You should test what happens if you remove this trick and compute second derivatives in a more direct way. Does training become unstable? Does accuracy change? This helps clarify whether the proposed method is essential or simply convenient.

You are also encouraged to explore more expressive quantum circuits. Many current implementations remain easy to simulate classically, which limits the potential for any genuine quantum advantage. Try increasing circuit depth, introducing more complex entangling structures, or otherwise pushing toward regimes that are harder to simulate. Even if this does not immediately improve performance, it helps probe where quantum models might become interesting.

Generalization is another key aspect that is often overlooked. After training your model, test it outside the conditions it was trained on. This could mean evaluating it beyond the original spatial or temporal domain, changing boundary conditions, or modifying the initial condition. The question here is whether the model has learned the underlying physics or is simply interpolating within the training region.

You can also explore resource usage: not just parameter counts, but a more physical notion of cost inspired by recent work on energetic analysis ([2601.08068] Quantum Energetic Advantage before Computational Advantage in Boson Sampling). If a quantum approach is more expensive to train or simulate, that should be part of the story.

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Finally, for teams with access to hardware, you may attempt to run the quantum circuit on an actual photonic processor. In this case, the model can still be trained classically, but inference is performed on the quantum device. This allows you to study the impact of noise and hardware imperfections in a realistic setting.

The goal is not to build a more complex model, but to better understand when – and if – the quantum component provides a meaningful advantage.

Tools and Resources

To help you get started quickly, we provide a set of baseline notebooks:

- *A MerLin starter notebook, which introduces the photonic quantum layer and basic usage within PyTorch.*
- *A classical ML PINN notebook, which implements a standard physics-informed neural network without any quantum component. This is especially useful as a baseline for benchmarking and ablation studies.*

You are encouraged to build on these resources, adapt them, and extend them as needed for your experiments.

For teams interested in exploring realistic execution, models can also be run on Quandela's photonic cloud platform, allowing you to study the impact of noise and hardware constraints in practice.

***Quandela Hub:** Cloud access – just click "Get Started For Free" and set up an account!*

Deliverables

By the end of the challenge, each team should provide:

- *A working implementation of the QPINN model using MerLin*
- *A clear comparison with at least one classical baseline*
- *Quantitative results (e.g. relative L^2 error, PDE residuals)*
- *Visualizations of predictions versus ground truth*
- *A short presentation summarizing your approach, experiments, and conclusions*

The most important part of the deliverable is not just the code, but the clarity of your analysis and the strength of your conclusions.

***Upload:** Challenger Folder – Solution Uploads*

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Evaluation Criteria

Projects will be evaluated based on:

- ***Rigor of investigation:*** *quality of experimental design, fairness of comparisons, and depth of benchmarking and ablations used to support conclusions*
- ***Quantum insight and validity:*** *how clearly you assess whether the quantum component provides a meaningful contribution, including analysis of when and why it helps (or does not), and whether the model operates in a genuinely quantum regime*
- ***Technical execution:*** *correctness, stability, and completeness of your implementation, including the quality of both the quantum and classical baselines and the overall experimental setup*
- ***Clarity of conclusions and communication:*** *how clearly and convincingly you present your findings, explain observed behaviors, and support your conclusions with evidence*

Strong projects are those that provide convincing evidence and thoughtful interpretation, regardless of whether the quantum model outperforms classical baselines.

Support

Throughout the hackathon, ***Quandela Knights will be on site****. They can assist with:*

- *understanding MerLin and photonic quantum models*
- *debugging technical issues*
- *discussing experimental design and benchmarking strategies*
- *support through a direct contact at Quandela*

Don't hesitate to ask questions – discussions are part of the process!

What matters

This challenge is not about building the most complex quantum circuit. It is about understanding what the model is doing and being able to support your conclusions with evidence.

A strong project might show that the quantum layer does not outperform a classical model under fair conditions. Another might find that the auxiliary derivative improves stability but not accuracy. Another might identify regimes where a hybrid model is more sample-efficient.

What matters most is that you treat the problem rigorously and answer, as clearly as possible:

Does the quantum component help, and if so, why?